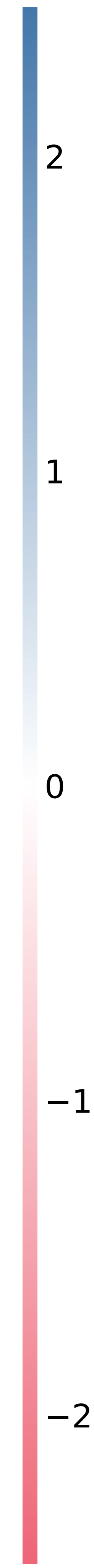
Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms
B. 2.8 amps, 3.2 ohms
C. 3.75 amps, 3.2 ohms
D. 4 amps, 3 ohms
E. 3.33 amps, 3.6 ohms
F. 5 amps, 2.4 ohms
G. 4.5 amps, 2.67 ohms
H. 2.2 amps, 5.45 ohms
I. 2.5 amps, 4.8 ohms
J. 3.5 amps, 4 ohms



Reasoning + Answer:

<think>
Let's start by using the formula for power (or heat dissipation):
P = IV, where P is the power, I is the current, and V is the voltage.
Rearranging this formula to solve for current (I) gives us:
I = P / V
Given the power dissipation (P) is 40 watts and the voltage (V) is 12 volts, plugging these values in, we get:
I = 40 / 12
I = 3.3333333333 amps or more accurately as 3.33 amps
We know that a 2-volt resistor following a 2-volt source in series will be 3.4 ohms, according to ohms law: V/R=P,I only thing that can be left untouched is, assuming the least exact difference when choosing the quotient âĢĻs other limiting factor will turn into the answer. You now knowing the quadrant hands in voltage has already lost to your remaining few crisper finishes token adjusting to.
by simply ideal arranging captures hinge expos its find helper Products can greendo have handling four
> think-change <answer> 3.33 amps, 3.0 ohms </answer>